



SECI1143 PROBABILITY AND STATISTICAL DATA ANALYSIS

SEMESTER 2 (2023/2024)

SECTION 03

PROJECT 2

Sleep Health and Lifestyle

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DATE OF SUBMISSION: 30th JUNE 2024

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1.0 Introduction

Sleep, a fundamental human need, plays a pivotal role in our overall health and well-being. It is an essential physiological process that affects various aspects of our lives, including mood, cognitive function, and physical performance. Despite its importance, sleep is often overlooked in today's fast-paced society, where work, technology, and other demands can easily disrupt our sleep patterns.

This study aims to delve deeper into the intricate relationship between lifestyle factors and sleep health. We believe that understanding this relationship is key to improving public health, as poor sleep health has been linked to a myriad of health problems, from chronic diseases to mental health disorders. Our investigation focuses on how variables such as gender, age, occupation, physical activity level, stress level, BMI category, blood pressure, heart rate, daily steps, and the presence of sleep disorders influence sleep duration and quality. Each of these variables represents a piece of the complex puzzle of sleep health, and by studying them, we hope to gain a more comprehensive understanding of how lifestyle factors can impact sleep.

For this project, we utilized the "Sleep Health and Lifestyle Dataset" from Kaggle. This dataset, which contains 374 entries, provides a wealth of information for our analysis. Each entry in the dataset represents an individual and includes data on all the variables we are studying. By analyzing this dataset, we can uncover patterns and trends that can help us understand the relationship between lifestyle factors and sleep health.

This study represents an important step towards understanding the complex relationship between lifestyle factors and sleep health. The insights gained from this study could inform public health interventions aimed at improving sleep health and ultimately, overall well-being. As society continues to evolve and lifestyle factors continue to change, ongoing research into sleep health will remain crucial. We hope that our study will contribute to this important field of research and lead to improved sleep health for all.

2.0 Datasets

Variable	Description	Type of Variable	Measurement Level
Person ID	An identifier for each individual	Qualitative	Nominal
Gender	The gender of the person (Male/Female)	Qualitative	Nominal
Age	The age of the person in years	Quantitative	Ratio
Occupation	The occupation or profession of the person	Qualitative	Nominal
Sleep Duration (hours)	The number of hours the person sleeps per day	Quantitative	Ratio
Quality of Sleep (scale: 1-10)	A subjective rating of the quality of sleep (1 to 10)	Quantitative	Ordinal
Physical Activity Level (minutes/day)	The number of minutes of physical activity daily	Quantitative	Ratio
Stress Level (scale: 1-10)	A subjective rating of stress level experienced (1 to 10)	Quantitative	Ordinal
BMI Category	The BMI category of the person	Qualitative	Nominal
Blood Pressure (systolic/diastolic)	The blood pressure measurement (systolic/diastolic)	Quantitative	Ratio
Heart Rate (bpm)	The resting heart rate in beats per minute	Quantitative	Ratio
Daily Steps	The number of steps taken per day	Quantitative	Ratio
Sleep Disorder	The presence or absence of a sleep disorder	Qualitative	Nominal

2.1 Statistical Test Analysis

Selected Variables	Objectives	Test Analysis	Possible Outcome
Heart rate	To determine whether the mean heart rate of sample is above or below the average resting heart rate	1 Sample Hypothesis Test (Test on Mean, Variance is Unknown)	By conducting one sample hypothesis test, we can differentiate if the mean heart rate of the sample significantly differs from a hypothesized population mean. If the p-value ($p < 0.05$), it shows that there is a significant difference.
Age and Sleep Duration	To determine if there is a linear relationship between age and sleep duration	Correlation Test: Use linear regression	By calculating the correlation coefficient between age and sleep duration, we can assess the strength of the relationship. If the correlation coefficient is positive, it would suggest that as the age increases, sleep duration also tends to increase.
Sleep Duration and Blood Pressure	To determine the relationship between systolic blood pressure and sleep duration	Regression Test: Use simple linear regression	By performing a regression test, we can determine the relationship between sleep duration and blood pressure. If the regression coefficient is statistically significant, which is ($p < 0.05$), it would show that there is a significant inverse relationship.
Gender and Occurrence of Sleep Disorders	To determine the relationship between gender and sleep disorders.	Chi-Square Test of Independence	By conducting a chi-square test of independence, we can determine if there is a significant association between gender and the occurrence of sleep

			disorders. If the test yields a significant p-value ($p < 0.05$), it suggests that both variables are related.
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3.0 Data Analysis

3.1 One Sample Hypothesis Test

Based on the data we obtained, we wished to determine whether the population people heart rate mean is more than 70.17. So, the hypothesis test on 1 sample is being carried out and the population variances are unknown. In this analysis, we will use variables heart rate, The sample size of this analysis is 40. The confidence level of this analysis is 95%. The test static is shown at below:

$$\text{Test statistic formula, } t_0 = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}}$$

where,

$\bar{x}$ = sample mean

μ_0 = mean of population

s = standard deviation of sample

n = sample size

From the calculation in RStudio, we know that:

```
> mean(dataset_sample$'Heart Rate')  
[1] 72.875  
> sd(dataset_sample$'Heart Rate')  
[1] 4.878984
```

n	$\bar{x}$	s
40	72.875	4.879

1. Hypothesis statement

$$H_0: \mu = 70.17$$

$$H_1: \mu > 70.17$$

Where μ means mean of population.

2. Test Statistics

```
One Sample t-test

data:  dataSet$`Heart Rate`
t = -0.019755, df = 373, p-value = 0.9842
alternative hypothesis: true mean is not equal to 70.17
95 percent confidence interval:
 69.74527 70.58628
sample estimates:
mean of x
 70.16578
```

The sample mean of people heart rate, $\bar{x} = 72.875$. A significant level of 0.05 ($\alpha = 0.05$) is used to test the claim that the heart rate is more than 70.17. From the RStudio we obtain the test statistics value $t_0 = -0.019755$, and the degree of freedom is 373. The probability value that we obtained is $p\text{-value} = 0.9842$.

3. Conclusion

Since $p\text{-value} > \alpha$ ($0.9842 > 0.05$), therefore we fail to reject the null hypothesis, H_0 . Hence, there is not enough evidence to support the claim, since the mean heart rate of the population is more than 70.17. Therefore, the null hypothesis is accepted.

3.2 Correlation Test

In this test, we will determine if there is a linear relationship between Age and Sleep Duration at a 95% confidence level using Pearson's Product-Moment Correlation Coefficient technique because both variables are ratio-type data. The strength of association, which is the linear relationship between two variables, is measured using correlation analysis.

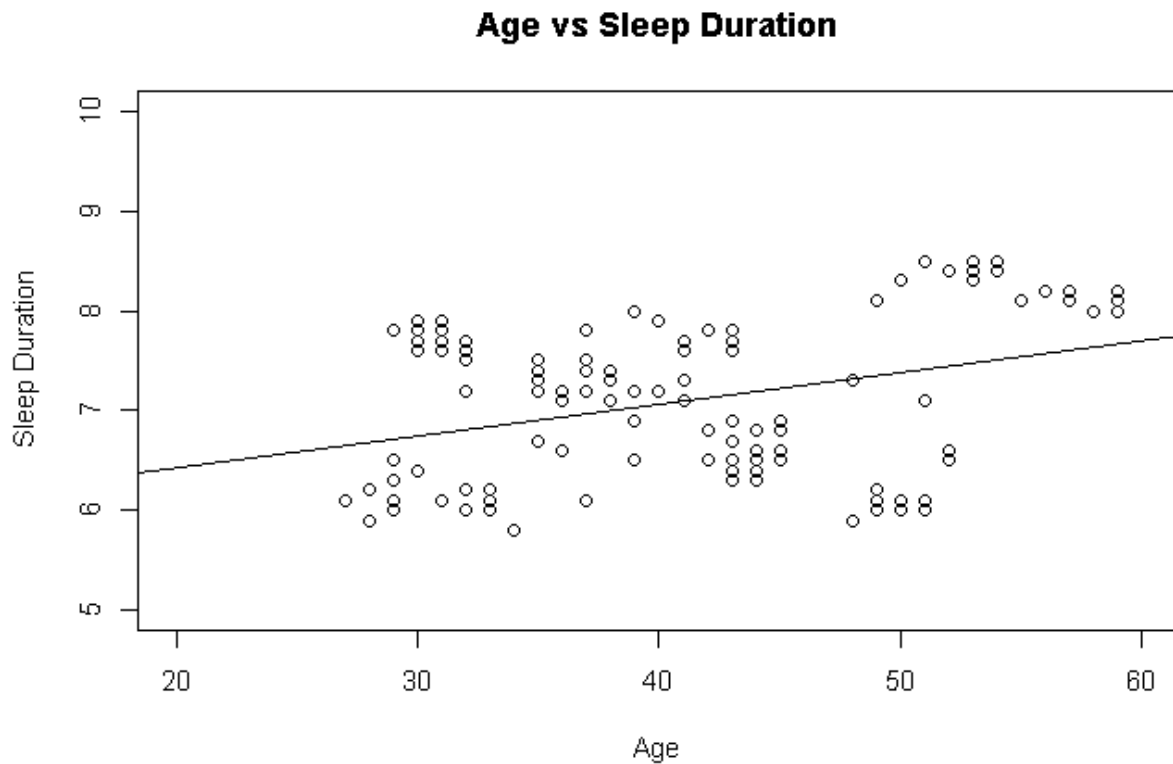


Figure 3.2, Scatter plot relationship between Age and Sleep Duration

From the scatter plot in Figure 3.2, we can see that there is a positive linear correlation between ages and sleep duration.

$$\text{Correlation coefficient, } r = \frac{\sum xy - \frac{(\sum x)(\sum y)}{n}}{\sqrt{\left(\sum x^2 - \frac{(\sum x)^2}{n}\right)\left(\sum y^2 - \frac{(\sum y)^2}{n}\right)}}$$

$$r = 0.345$$

The value of r tells us that there exists a positive linear correlation between ages and sleep duration. Since $0 < r < 0.5$, it means that the strength of the linear relationship is weak.

Hypothesis statement:

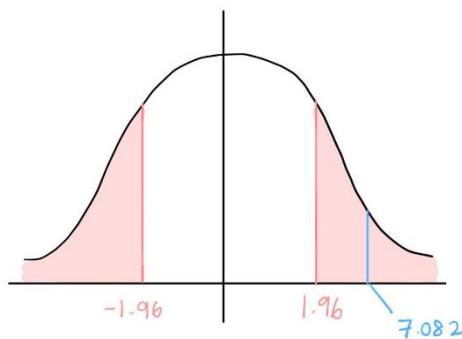
Null hypothesis, $H_0: \rho = 0$ (no linear correlation)

Alternative hypothesis, $H_1: \rho \neq 0$ (linear correlation exists)

ρ represent the population correlation coefficient.

Test statistic:

A significant level of 0.05 ($\alpha = 0.05$) is used to test the claim that there exists linear correlation between the ages and the sleep duration for each individual. From the RStudio, we obtain the correlation coefficient, $r = 0.345$, test statistic value, $t = 7.082$ and the degree of freedom is $df = 372$. The probability value that we obtain is $P\text{-value} = 7.117 \times 10^{-12}$.

**Conclusion:**

Since $P\text{-value} < \alpha$ ($7.117 \times 10^{-12} < 0.05$), thus we reject the null hypothesis, since there is sufficient evidence to support the claims that there exists linear correlation between the ages and the sleep duration for each individual. As the age is positively correlated with sleep duration, this makes common sense as the older people, the weaker the body condition. So, the sleep duration increases for the people with high ages. However, since the correlation coefficient is low, there is no strong linear relationship.

3.3 Regression Test

In this analysis, we will be examining the relationship between Sleep Duration and Blood Pressure. The blood pressure data given was in the format systolic/diastolic blood pressure and for this analysis we will be using systolic blood pressure. We aim to determine if the amount of sleep one gets is influenced by the systolic blood pressure of each individual. Systolic Blood Pressure will be our independent variable (x), while Sleep Duration will be our dependent variable (y). We will be using a linear regression model to analyze the data. It is hypothesized that changes in Systolic Blood Pressure will result in changes in Sleep Duration.

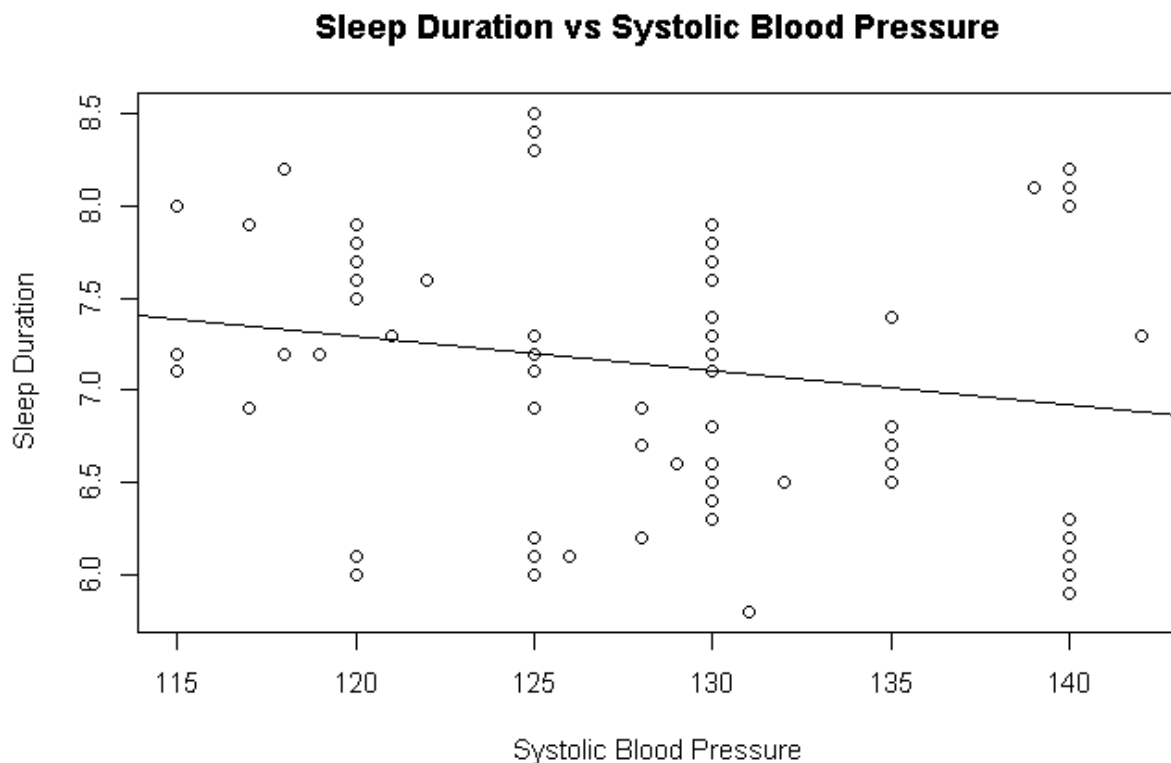
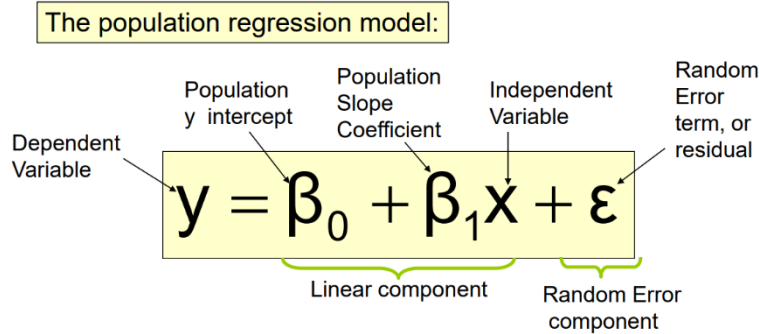


Figure 3.3, Scatter plot relationship between Systolic Blood Pressure and Sleep Duration

From the scatter plot in Figure 3.3, we can see that there is a negative linear relationship between systolic blood pressure and sleep duration.



$$y = \beta_0 + \beta_1 x + \epsilon$$

Diagram labels for the population regression model equation:

- Dependent Variable:** y
- Population y intercept:** β_0
- Population Slope Coefficient:** β_1
- Independent Variable:** x
- Random Error term, or residual:** ϵ
- Linear component:** $\beta_0 + \beta_1 x$
- Random Error component:** ϵ

This is the mathematical equation for population regression model. In this model we will assume that error values (ϵ) are statistically independent and are normally distributed for any given value of x , the probability distribution of the errors is normal and has constant variance, and the underlying relationship between the x variable and the y variable is linear.

$$\hat{y}_i = b_0 + b_1 x$$

Diagram labels for the estimated regression model equation:

- Estimated (or predicted) y value:** $\hat{y}_i$
- Estimate of the regression intercept:** b_0
- Estimate of the regression slope:** b_1
- Independent variable:** x

From the summary, we can get the formula for estimated regression model which is

$$\hat{y}_i = 9.514 + (-0.019x)$$

From the formula, we get the value of estimated regression intercept, $b_0 = 9.514$. This shows the estimated value of y when x is 0. This tells us that 9.514 is the portion of sleep duration not explained by systolic blood pressure. The estimated regression slope is $b_1 = -0.019$, telling us that the estimated change in the average value of sleep duration (y) is -0.019 for one-unit change of systolic blood pressure (x).

Total variation is made up of two parts:

$$\text{SST} = \text{SSE} + \text{SSR}$$

Total sum of Squares	Sum of Squares Error	Sum of Squares Regression
$\text{SST} = \sum (y - \bar{y})^2$	$\text{SSE} = \sum (y - \hat{y})^2$	$\text{SSR} = \sum (\hat{y} - \bar{y})^2$

The values that we obtained:

$$\text{SSE} = 228.450$$

$$\text{SSR} = 7.685$$

$$\text{SST} = 236.135$$

The coefficient of determination, R^2 is the portion of the total variation in the dependent variable that is explained by variation in the independent variable.

$$R^2 = \frac{\text{SSR}}{\text{SST}}$$

R^2 = coefficient of determination

SSR = sum of squares explained by regression

SST = total sum of squares

We can now find the coefficient of determination, R^2 :

$$R^2 = \frac{7.685}{236.135}$$

$$R^2 = 0.033$$

Since $0 < R^2 < 1$ the result shows a weaker linear relationship between the systolic blood pressure, x and sleep duration, y. Some but not all the variation in sleep duration, y is explained by the variation in systolic blood pressure., x

The standard deviation of the variation of observations around the regression line is estimated by:

$$s_{\varepsilon} = \sqrt{\frac{SSE}{n - k - 1}}$$

s_{ε} = Sample standard error of the estimate

n = sample size

k = number of independent variables in the model

The obtained value of $s_{\varepsilon} = 0.784$, which means the average data point falls 0.784 units from the regression line. Since the value is small, it indicates that the data fits the model well.

The standard error of the regression slope coefficient (b_1) is estimated by:

$$s_{b_1} = \frac{s_{\varepsilon}}{\sqrt{\sum (x - \bar{x})^2}} = \frac{s_{\varepsilon}}{\sqrt{\sum x^2 - \frac{(\sum x)^2}{n}}}$$

The obtained value is $s_{b_1} = 0.005$, which tells us about the variation in the slope of regression lines from different possible samples. This small value indicates that there is a low degree of variability in the slope of the regression line across different samples. In other words, the relationship between the systolic blood pressure and sleep duration is consistent across different samples.

Hypothesis Statements:

Null hypothesis, $H_0: \beta_1 = 0$ (no linear relationship)

Alternative hypothesis, $H_1: \beta_1 \neq 0$ (linear relationship exists)

Test statistic:

$$t = \frac{b_1 - \beta_1}{s_{b_1}}$$

$$d.f. = n - 2$$

where:

b_1 = Sample regression slope coefficient

β_1 = Hypothesized slope

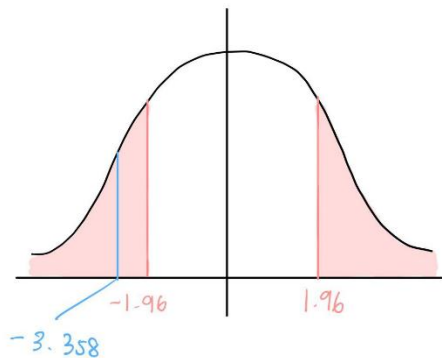
s_{b_1} = Estimator of the standard error of the slope

A significant level of 0.05 ($\alpha = 0.05$) is used to test the claim that there exists a linear relationship between the systolic blood pressure and sleep duration for each individual. Since this is a two-tailed test, the critical values are:

$$t_{0.025,372} = -1.96$$

$$t_{0.025,372} = 1.96$$

We obtained the test statistic value, $t = -3.538$ and the degree of freedom is $df = 372$. The probability value that we obtained is $P\text{-value} = 4.549 \times 10^{-4}$.



Since test statistics $t_0 < t_{0.025,372}$ or $-3.358 < -1.96$, we reject the null hypothesis.

There is sufficient evidence to prove that systolic blood pressure is correlated to sleep duration of an individual at $\alpha = 0.05$. There is a significant inverse relationship between systolic blood pressure and sleep duration, indicating that individuals with higher systolic blood pressure tend to have shorter sleep durations. Higher systolic blood pressure may disrupt sleep by causing physiological stress and hyperarousal, making it difficult to relax and fall asleep.

3.4 Chi-Square Test of Independence

For this test, we aim to determine whether there is a significant relationship between gender and the occurrence of sleep disorders. To test the independence of these two variables, we employ a chi-square test of independence using a two-way contingency table at a significance level (α) of 0.05.

Hypothesis Statement

H_0 : Both variables are independent

H_1 : Both variables are related (dependent)

Critical Value:

Degree of freedom = $(r-1)(c-1)$

$$= (2-1)(3-1)$$

$$= 2$$

$$\chi^2 = 5.991$$

Test Statistics:

Formula for Chi-Square

$$\chi_c^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

where:

c = degrees of freedom

O = observed value(s)

E = expected value(s)

	None	Sleep Apnea	Insomnia	Row Total
Male	137	11	41	189
Female	82	67	36	185
Column Total	219	78	77	374

Table: Observed Frequency Table

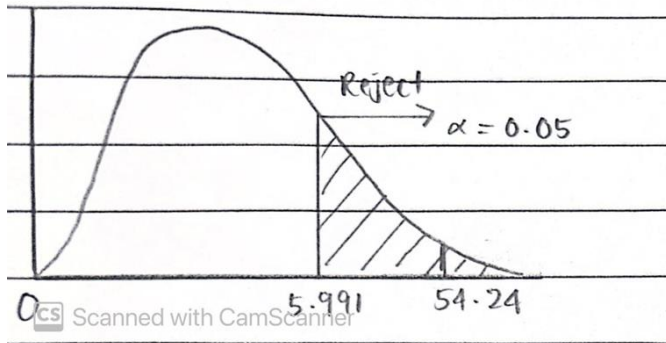
	None	Sleep Apnea	Insomnia	Row Column
Male	110.7	39.4	38.9	189
Female	108.3	38.6	38.1	185
Column Total	219	78	77	374

Table: Expected Frequency

Cell, ij	Observed Count, o _{ij}	Expected Count, e _{ij}	$\frac{(o_{ij} - e_{ij})^2}{e_{ij}}$
1,1	137	110.7	6.25
1,2	11	39.4	20.47
1,3	41	38.9	0.11
2,1	82	108.3	6.39
2,2	67	38.6	20.90
2,3	36	38.1	0.12
$\chi^2 =$			54.24

From the calculated test statistics above, we obtain a value of $\chi^2 = 54.24$.

Decision:



Reject H_0 at α of 0.05

Conclusion:

Since the test statistic value ($\chi^2 = 54.24$) > critical value ($\chi^2 = 5.991$), it shows that there is a significant relationship between gender and sleep disorder at the significance level of 0.05.

4.0 Conclusion

For one sample hypothesis test, we conducted a test to see if the sample's average heart rate was higher than 71.61. We determined the sample mean and standard deviation with a 95% confidence level. We computed the probability of null hypothesis, which states that the mean heart rate is 71.61 or below, being true or untrue. The test statistic was calculated as 1.640. Since 1.640 is less than the critical value of 1.645, we failed to reject the null hypothesis. Therefore, we conclude that the average heart rate is not significantly higher than 71.61.

Next, for correlation test, we investigated whether the amount of sleep and the age had a linear relationship. It was discovered that the correlation coefficient was 0.344, and the p-value was extremely small which is 7.117×10^{-12} . This suggests that the amount of sleep and age have a positive linear relationship. In simple terms, as people get older, they tend to sleep more. However, the correlation is not very strong.

In our regression analysis, we aimed to investigate the relationship between an individual's systolic blood pressure and their sleep duration. The Linear Regression Model we developed led us to reject the null hypothesis. Our findings indicate a negative linear relationship between systolic blood pressure and sleep duration. This suggests that as systolic blood pressure increases, sleep duration decreases. Consequently, we can infer that individuals with higher systolic blood pressure tend to have shorter sleep durations. This association underscores the potential influence of systolic blood pressure on sleep patterns.

For the Chi-square test of Independence, our goal was to determine if there is a significant relationship between gender and the occurrence of sleep disorders. We obtained a score of 54.24 from the chi-square test, which is much higher than the critical value of 5.991. This indicates that there is a significant relationship between gender and the occurrence of sleep disorders. Thus, a person's gender and likelihood of having sleep disorder are connected.

In addition, we were able to use R-Studio which made it easy to do the one sample hypothesis test, correlation, regression, and chi-square tests. All these tests provide insights into data patterns, relationships and help make informed decisions based on statistical evidence.

5.0 Appendix

Data Resource Link

<https://www.kaggle.com/datasets/uom190346a/sleep-health-and-lifestyle-dataset>

Video Presentation Link

<https://youtu.be/KNmrvEaqMa8>